DEM 5283/7283 Course Topics

Statistics for Demographic Data II, Spring 2027

Revised calendar

Week	Topic
1	Course introduction; reproducible workflow with Quarto, renv, and Git
2	Complex survey design, weights, replicate variance
3	Binary outcomes: logit and probit, predicted probabilities, AMEs
4	Model comparison across nested logits (KHB); prediction and cross validation
5	Ordinal and multinomial models
6	Counts and rates: Poisson, negative binomial, offsets, zero inflation
7	Decomposition: Kitagawa and Oaxaca-Blinder
8	Spring break
9	Missing data and multiple imputation with survey designs
10	Longitudinal models: GEE, fixed effects, within-between models
11	Splines and GAMs
12	Multilevel models I: centering, random slopes, level-specific weights
13	Multilevel models II: GLMMs, growth curves, APC identification
14	Causal inference foundations: DAGs and propensity score weighting
15	Tree-based machine learning with tidymodels
16	Final papers due